

Synthesis of Sensor Attacks for Tampering Detectability of Partially Observed Discrete-Event Systems

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Abstract—This article addresses the problem of the detectability of partially observed discrete-event systems subject to covert sensor attacks. We assume that an attacker can arbitrarily alter sensor readings after intercepting them from a target system, aiming to trick a predefined supervisor to issue improper control commands, making the system not strongly or weakly detectable. The definitions of attackable strong detectability and attackable weak detectability are presented. Necessary and sufficient conditions for the two types of attackable detectability are then derived based on the observer construction. The attack against strong detectability (AASD) problem and attack against weak detectability (AAWD) problem are formulated. It is shown that the supremal (or least restrictive) AASD and AAWD exist and can be computed, as long as the plant model and the supervisor model are regular, i.e., both are representable by finite-state automata.

Index Terms—Cyber security, detectability, discrete-event system (DESS), partial observation, sensor attack, supervisory control.

I. INTRODUCTION

STATE estimation is paramount for discrete-event systems (DESSs) due to their partial observations, as it serves as a critical step for supervisory control, security property analysis and verification, fault diagnosis and diagnosability analysis, and so forth. Several applications of state estimation in medical systems have been reported in, e.g., [7] and [8], where it is

essential to know the state of a system representing, for example, the severity stage of a patient's illness, before a treatment plan for the patient is made and issued. Another application of the state estimation problem is for a central station to be able to determine the state of a remote system under limited communications. As one specific measure of the system's ability of state estimation, detectability is used to study if an observation is sufficient to determine the current state for a given system. There are two types of detectability in a system, as defined in [4] and [5], namely, detectability where the current and subsequent states of the system can be inferred after a finite number of observations and periodic detectability where the state of the system can be periodically obtained. There are different variations of detectability in the community of DESSs. In this article, we restrict our attention to the design of a smart sensor attack for tampering two existing notions pertaining to detectability in DESSs: strong and weak detectability properties.

There have been a growing number of documented results addressing the cyber security issues from both the computer science community and the systems control community within the framework of continuous-variable dynamic systems and DESSs; the latter focuses on system dynamics affected by cyber attacks [13], [14], [15], [16], [22] and the supervisor synthesis against attacks [17], [18], [19], [20], [21]. Recently, more efforts have been made in classifying distinct types of malicious attacks, assuming that the attackers are sufficiently smart [10], [11], instead of merely causing random corruptions to system behavior, which has been studied in the fields of reliability and fault tolerant control. Typically, a smart attacker requires *system knowledge*, and abilities for *resource disclosure* and *resource disruption* in order to carry out a successful attack, which is covert to a system user until the attacker's goal of causing damage or corruptions to a system is achieved [1]. Tampering the strong detectability and weak detectability of a system is treated as the target of an attacker in this work. In this sense, covertness and damage infliction are two major characteristics of a successful attack.

In this article, we touch upon a special type of data deception attack in the framework of DESSs, where an attacker can intercept sensor measurements (or observations) modeled by observable events and alter them arbitrarily but with an upper bound imposed on the length of each altered observation sequence, following and extending our recent work in [12]. The general schematic

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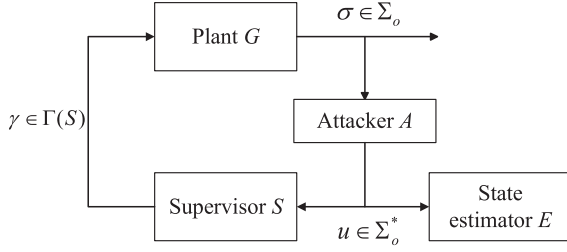


Fig. 1. General architecture of a closed-loop system under a sensor attack.

of the problem setting in this research is depicted in Fig. 1, where G is a plant, S plays the role of a supervisor and an anomaly detector in the closed-loop system (G, S) together with a state estimator E , and $\Gamma(S)$ is the set of control commands generated by S . If the state estimator can strongly (respectively weakly) determine the current state of (G, S) based on observations, then the closed-loop system is said to be strongly (respectively weakly) detectable. A sensor attacker A is supposed to engage in the system between the output of the plant G , and the input of the supervisor S and the state estimator E .

By sending the altered observation sequences to the supervisor whose control logic is known to the attacker in advance, the attacker is able to trigger the state estimator such that the latter makes incorrect state estimations, which is not perceptible by the supervisor's anomaly detection mechanism, as detailed shortly. To formalize this problem, we first explore two new properties of the closed-loop system under sensor attacks, namely, attackable strong detectability and attackable weak detectability. The former (respectively latter) describes whether or not a concerned closed-loop system under sensor attacks is strongly (respectively weakly) detectable [12]. Then, the concepts of *attack against strong detectability* (AASD) and *attack against weak detectability* (AAWD) are presented, which can be modeled as finite-state automata, possessing the properties of covertness, strong detectability damage, weak detectability damage, control feasibility under partial observations, and liveness. Finally, we show that there exist the supernal AASD and supernal AAWD that are computable via specific synthesis procedures, as long as both the plant model G and the given supervisor S are finitely representable (admitting a finite state automaton representation).

The construction of a sensor attack model as a transducer-style automaton is inspired by a recent work on the smart sensor attack [1], which aims to design a smart sensor attack that is covert to a system and leads the system to move into an illegal state. Note that the goal of the attack design in [1] is different from this article. In this work, tampering the strong detectability and weak detectability of a system is treated as the target of an attacker. The main contributions of this research are summarized as follows.

- 1) We formulate the definitions of attackable strong detectability and attackable weak detectability. Necessary and sufficient conditions for the verification of attackable strong detectability and attackable weak detectability in the considered closed-loop system under attacks are presented.

- 2) The synthesis problems for AASD and AAWD are defined. It is shown that there exist the supernal AASD and supernal AAWD that are computable if both plant and supervisor are finitely representable.

The rest of this article is organized as follows. Section II provides preliminaries on the system model and the definitions of strong detectability and weak detectability adopted from [4] and [5], respectively. In Section III, the problem statement is described, together with the notions of attackable strong detectability and attackable weak detectability, where the AASD and AAWD synthesis problems are formulated. Section IV provides the main results verifying the two types of attackable detectability and several examples illustrating the proposed methods. Section V centers around the conditions under which the supernal AASD and supernal AAWD exist and are computable. Finally, Section VI concludes this article.

II. PRELIMINARY

This section first recalls the basic concepts of the Ramadge-Wonham's supervisory control paradigm [2] used throughout this article. Then, we review the notions of detectability of DESs in [4] and [5].

A. Supervisory Control

Let $\mathbb{R}^+$ be the collection of all positive reals and $\mathbb{N}$ be the set of natural numbers (containing zero). Let Σ be a finite alphabet, grouping the labels of events in a discrete-event system [2]. We use Σ^+ to denote the set of all finite sequences of symbols defined over Σ and use Σ^* to denote the Kleene closure of Σ , i.e., $\Sigma^* := \Sigma^+ \cup \{\epsilon\}$, where ϵ is the empty sequence. An element $s \in \Sigma^*$ is called a *string*. We use $\sigma \in s$ to denote that an event (label) σ appears at least once in s , and $s^\downarrow$ to denote the last event of s if $s \neq \epsilon$. Given two strings $s, t \in \Sigma^*$, s is called a *prefix substring* of t , written as $s \leq t$, if there exists $s' \in \Sigma^*$ such that $ss' = t$, where ss' denotes the concatenation of s and s' . For all strings $s \in \Sigma^*$, we have $\epsilon s = s\epsilon = s$. A subset $L \subseteq \Sigma^*$ is called a *language* over Σ . Given a language L , $\bar{L} = \{s \in \Sigma^* | (\exists t \in L) s \leq t\} \subseteq \Sigma^*$ is the *prefix closure* of L . Given two languages $L, L' \subseteq \Sigma^*$, $LL' = \{ss' \in \Sigma^* | s \in L \wedge s' \in L'\}$ is called the concatenation of L and L' .

Let $\Sigma' \subseteq \Sigma$ be a subset of Σ and $\text{Pwr}(\cdot)$ be the powerset of a set. The *natural projection* $P : \Sigma^* \rightarrow \Sigma'^*$ is defined by

$$\begin{cases} P(\epsilon) = \epsilon \\ P(\sigma) = \begin{cases} \sigma, & \text{if } \sigma \in \Sigma' \\ \epsilon, & \text{otherwise} \end{cases} \\ P(s\sigma) = P(s)P(\sigma), \text{ if } s \in \Sigma^* \text{ and } \sigma \in \Sigma. \end{cases} \quad (2-1)$$

The *inverse image function* of P is

$$P^{-1} : \text{Pwr}(\Sigma'^*) \rightarrow \text{Pwr}(\Sigma^*) \quad (2-2)$$

where for any $H \subseteq \Sigma'^*$

$$P^{-1}(H) := \{s \in \Sigma^* | P(s) \in H\}. \quad (2-3)$$

Given $L_1 \subseteq \Sigma_1^*$ and $L_2 \subseteq \Sigma_2^*$, the *synchronous product* of L_1 and L_2 is defined as

$$L_1 \| L_2 := P_1^{-1}(L_1) \cap P_2^{-1}(L_2) \quad (2-4)$$

where $P_i : (\Sigma_1 \cup \Sigma_2)^* \rightarrow \Sigma_i^* (i = 1, 2)$ are natural projections.

A given target plant is modeled as a *deterministic finite-state automaton* $G = (X, \Sigma, \xi, x_0, X_m)$, where X stands for the state set, Σ stands for the alphabet, $\xi : X \times \Sigma \rightarrow X$ stands for the (partial) transition function, x_0 stands for the initial state, and $X_m \subseteq X$ stands for the marker state set. Given $\sigma \in \Sigma$, following the notations in [2], we use $\xi(x, \sigma)!$ to denote that the transition $\xi(x, \sigma)$ is defined. For a state $x \in X$, let $En_G(x) := \{\sigma \in \Sigma | \xi(x, \sigma)!\}$ be the set of events enabled at state x in G . The domain of ξ can be extended to $X \times \Sigma^*$, where $\xi(x, \epsilon) = x$ for all $x \in X$ and $\xi(x, s\sigma) := \xi(\xi(x, s), \sigma)$, where $s \in \Sigma^*$ and $\sigma \in \Sigma$. The *closed* behavior of G is defined as $L(G) = \{s \in \Sigma^* | \xi(x_0, s)!\}$ and the *marked* behavior of G is $L_m(G) = \{s \in L(G) | \xi(x_0, s) \in X_m\}$. G is *nonblocking* if $\overline{L_m(G)} = L(G)$. We will use $|X|$ to denote the size of its state set and $|\Sigma|$ for that of Σ . Given two finite-state automata $G_i = (X_i, \Sigma, \xi_i, x_{i,0}, X_{i,m})$, $i = 1, 2$, their *meet*, denoted as $G_1 \wedge G_2$, is a (reachable) finite-state automaton whose alphabet is Σ such that $L(G_1 \wedge G_2) = L(G_1) \cap L(G_2)$ and $L_m(G_1 \wedge G_2) = L_m(G_1) \cap L_m(G_2)$.

We now recall the concept of supervisors. Let $\Sigma = \Sigma_c \dot{\cup} \Sigma_{uc} = \Sigma_o \dot{\cup} \Sigma_{uo}$, where Σ_c (Σ_o) and Σ_{uc} (Σ_{uo}) are disjoint, denoting, respectively, the sets of *controllable* (*observable*) and *uncontrollable* (*unobservable*) events. A (feasible) *supervisory control map* of G under partial observation $P_o : \Sigma^* \rightarrow \Sigma_o^*$ is defined as $V : L(G) \rightarrow 2^\Sigma$ such that the following holds.

- 1) $(\forall s \in L(G))(\forall \sigma \in \Sigma_{uc}) s\sigma \in L(G) \Rightarrow \sigma \in V(s)$.
- 2) $(\forall s, s' \in L(G)) P_o(s) = P_o(s') \Rightarrow V(s) = V(s')$.

For any $s \in L(G)$, $V(s)$ is interpreted as the set of events allowed to be enabled after event sequence s occurs. In other words, a supervisory control map does not disable any uncontrollable events but imposes the same control pattern for the strings that cannot be distinguished based on observations, i.e., the same control pattern is enforced for the observationally equivalent strings. Let V/G denote the closed-loop system of G under supervision of V . The generated language of V/G , denoted by $L(V/G)$ [2], is defined as the following.

- 1) $\epsilon \in L(V/G)$.
- 2) $(\forall s \in L(V/G))(\forall \sigma \in \Sigma) s\sigma \in L(V/G) \iff s\sigma \in L(G) \wedge \sigma \in V(s)$.
- 3) No others strings are in $L(V/G)$.

As for the marked language of closed-loop system V/G , we have $L_m(V/G) = L_m(G) \cap L(V/G)$. It is nonblocking if $L(V/G) = \overline{L_m(V/G)}$. Let $S = (Z, \Sigma, \tau, z_0, Z_m)$ with $Z_m = Z$ be a finite-state automaton that implements the supervision V such that the following holds.

- 1) $L(S \wedge G) = L(V/G)$ and $L_m(S \wedge G) = L_m(V/G)$.
- 2) $(\forall s, s' \in L(S)) P_o(s) = P_o(s') \Rightarrow \tau(z_0, s) = \tau(z_0, s')$.

The last condition implies that $V(s) = V(s')$ if $P_o(s) = P_o(s')$. It is shown in [2] that, as long as a language $K \subseteq L_m(G)$ is *controllable*, *observable*, and $L_m(G)$ -closed, there always exists a finitely representable supervisory control map V such that $L_m(V/G) = K$ and $L(V/G) = \overline{K}$. From now on, we assume that V/G is finitely representable by S , which is usually called a *supervisor*, and let us write a supervisor $S = (Z, \Sigma, \tau, z_0, Z)$ in place of $S = (Z, \Sigma, \tau, z_0, Z_m)$.

B. Detectability

For distinct state estimation problems, different assumptions on the initial state are usually made. In this article, we assume that the initial state x_0 is exactly known. Given a deterministic finite-state automaton $G = (X, \Sigma, \xi, x_0, X_m)$, if the current state estimate is $X' \subseteq X$, then the set of all possible states after observing $t \in \Sigma_o^*$ is defined by

$$R_G(X', t) = \{x \in X | (\exists x' \in X')(\exists s \in P_o^{-1}(\{t\})) \\ x = \xi(x', s)\}. \quad (2-5)$$

In particular, the unobservable reach of state set $X' \subseteq X$ is defined as

$$UR_G(X') = R_G(X', \epsilon). \quad (2-6)$$

The observer of system G can be constructed by replacing all unobservable events in G with the empty string ϵ and converting it into a deterministic automaton through subset construction [5], [9]. Let the observer of G be

$$G_{\text{obs}} = (Q, \Sigma_o, \delta, q_0) = Ac(2^X, \Sigma_o, \delta, UR_G(\{x_0\})) \quad (2-7)$$

where $Ac(\cdot)$ denotes the reachable part of an underlying automaton. Note that $q \in Q$ is a subset of $X (q \subseteq X)$. The transition function $\delta : Q \times \Sigma_o \rightarrow Q$ is defined, for $q \subseteq X$ and $\sigma \in \Sigma_o$, as

$$\delta(q, \sigma) = R_G(q, \sigma). \quad (2-8)$$

Recall that the finite-state automaton S implements the supervisory control map V . A closed-loop system can be equivalently written as S/G . Given a closed-loop system $S/G = (Z \times X, \Sigma, \tau \times \xi, (z_0, x_0), Z \times X_m)$ [2], if the current state estimate is $X' \subseteq X$, then the set of all possible states after observing $t \in \Sigma_o^*$ is defined by

$$R_{S/G}(X', t) = \{x \in X | (\exists (z', x') \in Z \times X') \\ (\exists s \in P_o^{-1}(\{t\}))(\exists z \in Z)(z, x) = \tau \times \xi((z', x'), s)\}. \quad (2-9)$$

In particular, the unobservable reach of state set $X' \subseteq X$ is defined as

$$UR_{S/G}(X') = R_{S/G}(X', \epsilon). \quad (2-10)$$

The observer construction method of the closed-loop system S/G is similar to that of G [9].

The observer of S/G can be constructed by replacing all unobservable events in S/G with the empty string and converting it into an automaton through subset construction. Let the observer of S/G be

$$G_{\text{obs}}^{S/G} = (Q_{S/G}, \Sigma_o, \delta_{S/G}, q_{S/G,0}) \\ = Ac(2^{X \times X}, \Sigma_o, \delta_{S/G}, UR_{S/G}(\{x_0\})) \quad (2-11)$$

where $Ac(\cdot)$ denotes the reachable part of an underlying automaton and $L(G_{\text{obs}}^{S/G}) \subseteq L(G_{\text{obs}})$. Note that $q \in Q_{S/G}$ is a subset of $X \times X (q \subseteq X \times X)$. The transition function $\delta_{S/G} : Q_{S/G} \times \Sigma_o \rightarrow Q_{S/G}$ is defined, for $q \subseteq X \times X$ and $\sigma \in \Sigma_o$, as $\delta_{S/G}(q, \sigma) = R_{S/G}(q, \sigma)$. Denote the set of states in $G_{\text{obs}}^{S/G}$ that contain a

singleton state in X of G as

$$Q_s = \{q \in Q_{S/G} \mid |q| = 1\}. \quad (2-12)$$

Denote the set of all loops (circuits) in $G_{\text{obs}}^{S/G}$ as

$$\text{Loop} = \{(q, u) \in Q_{S/G} \times \Sigma_o^* \mid |u| \geq 1 \wedge \delta_{S/G}(q, u) = q\}. \quad (2-13)$$

Four types of detectability are investigated in [4] and [5]: detectability, strong detectability, periodic detectability, and strong periodic detectability. In this work, only strong detectability and weak detectability are considered. We first recall the definitions of strong detectability and weak detectability [4] and [5]. As in [4] and [5], we make the following assumptions.

Assumption 1: A closed-loop system S/G is deadlock-free, i.e., $(\forall s \in L(S/G))(\exists \sigma \in \Sigma) s\sigma \in L(S/G)$.

Assumption 2: A plant G cannot generate an arbitrarily long unobservable sequence, i.e., $(\forall x \in X)(\forall s \in \Sigma^*) \xi(x, s) = x \Rightarrow P_o(s) \neq \epsilon$.

Assumption 3: A supervisor S can only disable observable events, i.e., $\Sigma_c \subseteq \Sigma_o$.

It is known [2] that under Assumption 3, a closed-loop language is controllable and observable if and only if it is controllable and normal. Imposing normality can guarantee that there exists a supremal (maximally permissive) supervisor model and bring some computational advantages. Moreover, Assumption 3 can be easily satisfied in reality, as it requires that only the events that are both observable and controllable should be disabled in real applications; it is typical that all control commands are observable.

Definition 1: Let $G = (X, \Sigma, \xi, x_0, X_m)$ be a plant and $S = (Z, \Sigma, \tau, z_0, Z)$ be its supervisor. A closed-loop system $S/G = (Z \times X, \Sigma, \tau \times \xi, (z_0, x_0), Z \times X_m)$ is strongly detectable with respect to P_o if

$$(\exists k \in \mathbb{N})(\forall s \in L(S/G)) |P_o(s)| \geq k \Rightarrow |R_{S/G}(q_{S/G_0}, P_o(s))| = 1 \quad (2-14)$$

where q_{S/G_0} is the initial state of the observer $G_{\text{obs}}^{S/G}$ of S/G .

Let $s \in L(S/G)$ be a sequence generated by a closed-loop system. Condition (2-14) indicates that there exists $k \in \mathbb{N}$ such that, once $|P_o(s)| \geq k$, the set of all possible states after observing $P_o(s)$ contains a singleton state. Alternately speaking, strong detectability describes that we can unambiguously determine, after a finite number of observations, the current state and subsequent states for all arbitrarily long trajectories of the system.

Definition 2: Let $G = (X, \Sigma, \xi, x_0, X_m)$ be a plant and $S = (Z, \Sigma, \tau, z_0, Z)$ be its supervisor. A closed-loop system $S/G = (Z \times X, \Sigma, \tau \times \xi, (z_0, x_0), Z \times X_m)$ is weakly detectable with respect to P_o if

$$(\exists k \in \mathbb{N})(\exists s \in L(S/G)) |P_o(s)| \geq k \wedge (\exists t \in \Sigma^*) [|P_o(t)| \geq |X| \wedge st \in L(S/G)] \Rightarrow (\forall t' \leq t) |R_{S/G}(q_{S/G_0}, P_o(st'))| = 1 \quad (2-15)$$

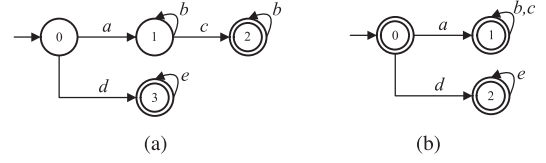


Fig. 2. Plant G and a supervisor S in Example 1. (a) Plant G . (b) Supervisor S .

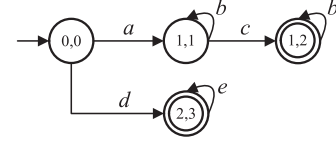


Fig. 3. Closed-loop system S/G in Example 1.

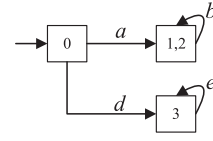


Fig. 4. Observer of S/G in Example 1.

where q_{S/G_0} is the initial state of the observer $G_{\text{obs}}^{S/G}$ of S/G . $\square$

The definition of weak detectability presented in this work is inspired by determining the current state and subsequent states for certain arbitrarily long trajectories of a closed-loop system in [5]. Suppose that there exists a sequence s generated by the closed-loop system. Condition (2-15) requires that there should exist $k \in \mathbb{N}$ such that, if $|P_o(s)| \geq k$, there is a sufficiently long continuation t of s with $|P_o(t)| \geq |X|$ and $st \in L(S/G)$, such that for any prefix substring t' of t , then the set of all possible states after observing $P_o(st')$ is a singleton. In other words, weak detectability describes a fact that we can determine, after a finite number of observations, the current state and subsequent states for certain (not all) arbitrarily long trajectories of the system.

Given a DES G , Shu and Lin [5] constructed an observer of G , concluding that G is strongly detectable if and only if any state reachable from any loop in the observer is of cardinality one; it is weakly detectable if and only if there is a loop in the observer consisting of singleton DES state sets. Note that a state in the observer of a plant G is a set of states of G .

Example 1: A plant G and its supervisor S are shown in Fig. 2, where $\Sigma_o = \Sigma_c = \{a, b, d, e\}$. A double-circled state denotes a marker state. The closed-loop system S/G is depicted in Fig. 3 and its observer $G_{\text{obs}}^{S/G}$ is portrayed in Fig. 4. After an arbitrarily long sequence ab^* is observed, the state of the plant is estimated to be x_1 or x_2 , implying that one cannot uniquely determine the current state and subsequent states. After the arbitrarily long sequence de^* is observed, the state of the plant is estimated to be x_3 , and thus the current state and subsequent states can be solely determined. We conclude that the system is not strongly detectable, but weakly detectable. Note that if all events are observable, i.e., $\Sigma_o = \{a, b, c, d, e\}$, the system is strongly detectable. $\square$

III. SMART SENSOR ATTACK FOR TAMPERING DETECTABILITY PROBLEM

In this work, we continue to use the sensor attack model proposed in [1], where an attacker is modeled by a transducer-style automaton.

A. Sensor Attack Model

Given a plant G with Σ_o being the set of observable events and its supervisor S , an attacker can intercept an observable event generated by G , and replace it with a sequence of observable events from Σ_o^* in order to trick the supervisor S whose function is known to the attacker. Here, we briefly recall the sensor attack model developed in [1].

Considering that in practice any event occurrence takes a nonnegligible amount of time, it is impossible for an attacker to insert an arbitrarily long observable event sequence to replace a received observable event. To this end, we assume that there exists a known natural number $n \in \mathbb{N}$ such that the length of any observable sequence that the attacker can insert is not greater than n . Let $\Delta_n = \{s \in \Sigma_o^* \mid |s| \leq n\}$ be the set of all such length-bounded observable sequences, where $|s|$ denotes the length of s . A sensor attack is modeled as a finite-state automaton $A = (Y, \Sigma \times \Delta_n, \eta, y_0, Y_m)$, where Y is the state set, Σ is the input alphabet, Δ_n is the output alphabet, y_0 is the initial state, Y_m is the marker state set, which is particularly specified as $Y_m = Y$, and $\eta : Y \times \Sigma \times \Delta_n \rightarrow Y$ is the (partial) transition map, where $\eta(y, \sigma, \epsilon) = y$ for all $\sigma \in \Sigma_{uo}$ and $y \in Y$, i.e., at each state y all unobservable events are self-looped with ϵ as the output. Let $\psi : (\Sigma \times \Delta_n)^* \rightarrow \Sigma^*$ and $\theta : (\Sigma \times \Delta_n)^* \rightarrow \Delta_n^*$ be the *input* and *output* maps, respectively, such that for a sequence $\mu = (\sigma_1, u_1)(\sigma_2, u_2) \cdots (\sigma_l, u_l) \in (\Sigma \times \Delta_n)^*$, we have $\psi(\mu) = \sigma_1 \sigma_2 \cdots \sigma_l$ and $\theta(\mu) = u_1 u_2 \cdots u_l$. Through input map ψ and output map θ , the unaltered observable event inputs received by the attacker and the observable event outputs inserted by the attacker can be obtained, respectively. We use $\mu^\dagger$ to denote the last event (σ_l, u_l) of $\mu = (\sigma_1, u_1)(\sigma_2, u_2) \cdots (\sigma_l, u_l)$. From now on, let us write a sensor attack $A = (Y, \Sigma \times \Delta_n, \eta, y_0, Y)$ in place of $A = (Y, \Sigma \times \Delta_n, \eta, y_0, Y_m)$.

We recall the attack procedure, as depicted in Fig. 1. If a sensor attack intercepts the plant output and modifies its content, then it may mislead the supervisor to move to a state that should be forbidden by the supervisor, without being noticed by the supervisor. In this case, the state estimation may be changed due to the corrupted observation. Specifically, the observer of (G, S) can be used to estimate the state of the original closed-loop system without attacks. However, the state estimations due to the observer of (G, S) are in general no longer correct once the closed-loop system is subject to attacks. To this end, it is necessary to construct an observer to correctly make state estimations under attacks.

The sequential composition of the attack model A and the supervisor S forms a new supervisor, denoted as $S \circ A$, which, for an observed output $\sigma \in \Sigma_o$, generates a control command $\gamma \in \Gamma(S) := \{V(s) \mid s \in L(S)\}$. Given a supervisor $S = (Z, \Sigma, \tau, z_0, Z)$ and a sensor attack $A = (Y, \Sigma \times \Delta_n, \eta, y_0, Y)$, their *sequential composition* [1] is a

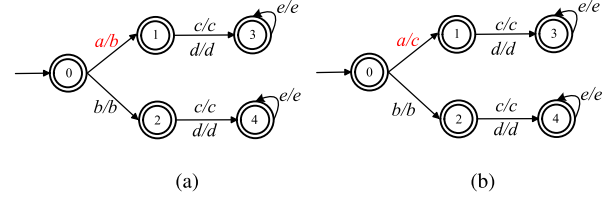


Fig. 5. Attackers A_1 and A_2 . (a) Attacker A_1 . (b) Attacker A_2 .

deterministic finite-state automaton $S \circ A = (Z \times Y \cup \{d\}, \Sigma \times \Delta_n, \zeta, (z_0, y_0), Z \times Y \cup \{d\})$, where d is an additionally introduced dump state, and for all $w \in Z \times Y \cup \{d\}$ and $(\sigma, v) \in \Sigma \times \Delta_n$

$$\zeta(w, \sigma, v) := \begin{cases} (z', y'), & \text{if } w = (z, y), \eta(y, \sigma, v) = y' \\ & \text{and } \tau(z, v) = z' \\ d, & \text{if } w = (z, y), \eta(y, \sigma, v) = y' \\ & \tau(z, \sigma)!, \text{ and } \tau(z, v) \text{ undefined} \\ \text{undefined,} & \text{otherwise.} \end{cases}$$

The closed-loop system under attack, denoted by $(S \circ A)/G$, is represented by the following finite-state automaton [1]:

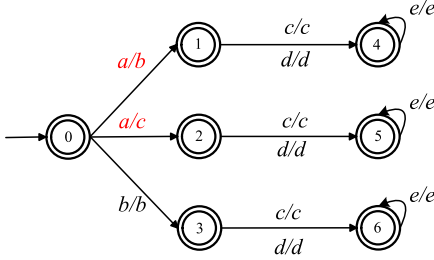
$(S \circ A)/G = ((Z \times Y \cup \{d\}) \times X, \Sigma \times \Delta_n, \kappa, (z_0, y_0, x_0), Z \times Y \times X_m)$ where for all $(w, x) \in (Z \times Y \cup \{d\}) \times X$ and all $(\sigma, v) \in \Sigma \times \Delta_n$

$$\kappa(w, x, \sigma, v) := \begin{cases} (w', x'), & \text{if } \xi(x, \sigma) = x' \text{ and} \\ & w' = \zeta(w, \sigma, v) \\ \text{undefined,} & \text{otherwise.} \end{cases}$$

Let $\Sigma_{o,p} \subseteq \Sigma_o$ denote the set of *protected* observable events in a DES that cannot be corrupted by a sensor attack [1], i.e., given an attack model $A = (Y, \Sigma \times \Delta_n, \eta, y_0, Y)$, the following predicate holds:

$$(\forall y \in Y)(\forall (\sigma, v) \in \Sigma_{o,p} \times \Delta_n) \eta(y, \sigma, v)! \Rightarrow v = \sigma. \quad (3-1)$$

Given two attackers A_1 and A_2 with the same input alphabet Σ and output alphabet Δ_n , by a slight abuse of notation, write $A_1 \cup A_2$ to denote a deterministic finite-state automaton, resulting from the *union* of A_1 and A_2 [1] such that $L(A_1 \cup A_2) = L(A_1) \cup L(A_2)$, as illustrated through a simple example. Suppose that A_1 and A_2 are attackers, as depicted in Fig. 5, where $\Sigma_o \setminus \Sigma_{o,p} = \Sigma_o = \Sigma = \{a, b, c, d, e\}$. In the attack model, a transition label is represented by a pair of events σ_1/σ_2 with $\sigma_1, \sigma_2 \in \Sigma_o$, where σ_1 is an observable event received from the plant G , and σ_2 is a faked observable event generated by the attacker, after intercepting event σ_1 . In the model of attacker A_1 , the attacker replaces event a with b , which is marked in red. In the model of attacker A_2 , the attacker replaces event a with c , which is also marked in red. The union of A_1 and A_2 obtained by simply merging their initial states is shown in Fig. 6, which is a deterministic finite-state automaton representing a nondeterministic attack. In other words, one of the two situations a/b and a/c will be nondeterministically chosen, i.e., the supervisor and external state estimator can see either b or c . In the rest of this article, the following assumption is also made.

Fig. 6. Union of A_1 and A_2 .

Assumption 4: The system state estimator is unaware of the existence of a sensor attack.

Given a closed-loop system under attack $S \circ A/G = ((Z \times Y \cup \{d\}) \times X, \Sigma \times \Delta_n, \kappa, (z_0, y_0, x_0), Z \times Y \times X_m)$, its observer G_{obs}^A can be constructed by replacing all unobservable events in $S \circ A/G$ with the empty string and converting it into an automaton through subset construction. Define G_{obs}^A , where the marker states are of no interest, as

$$G_{\text{obs}}^A = (Q', \Delta_n, \delta', q'_0) = Ac(2^X, \Delta_n, \delta', UR_{S/G}(\{x_0\})) \quad (3-2)$$

where $Ac(\cdot)$ denotes again the reachable part of an underlying automaton and $L(G_{\text{obs}}^A) \subseteq L(G_{\text{obs}})$. As defined, a state q' in Q' is a subset of X , i.e., $q' \subseteq X$. The transition function $\delta' : Q' \times \Delta_n \rightarrow Q'$ is defined, for $q' \subseteq X$ and $v \in \Delta_n$, as

$$\delta'(q', v) = R_{S \circ A/G}(q', v) \quad (3-3)$$

where $R_{S \circ A/G}(q', v) :=$

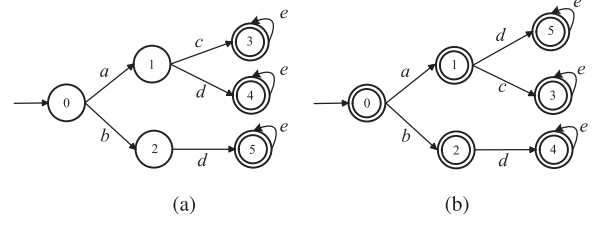
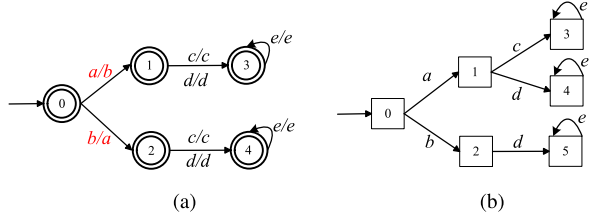
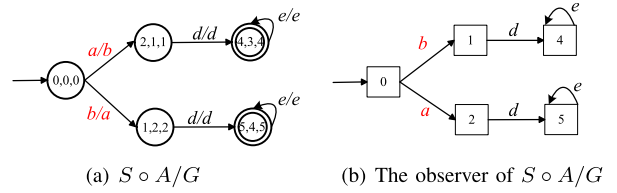
$$\begin{aligned} & \{x \in X | (\exists z', y', x') \in (Z \times Y \cup \{d\}) \times q' \\ & (\exists \mu \in (\Sigma \times \Delta_n)^*) (\exists (z, y) \in Z \times Y) \\ & \theta(\mu) = v \wedge (z, y, x) = \kappa((z', y', x'), \mu)\}. \end{aligned} \quad (3-4)$$

Definition 3: Given a plant G , a supervisor S , and a deterministic attacker A , the closed-loop system under attack $S \circ A/G = ((Z \times Y \cup \{d\}) \times X, \Sigma \times \Delta_n, \kappa, (z_0, y_0, x_0), Z \times Y \times X_m)$ is said to be attackable with respect to strong detectability and Σ_o , if S/G is strongly detectable and the following predicate is true:

$$\begin{aligned} & (\forall k \in \mathbb{N}) (\exists \mu \in L(S \circ A/G)) |\theta(\mu)| \geq k \wedge \theta(\mu) \in \\ & P_o(L(S/G)) \Rightarrow [|R_{S \circ A/G}(q'_0, \theta(\mu))| > 1 \vee \\ & \{\xi(x_0, \psi(\mu))\} \neq R_{S/G}(q_{S/G_0}, \theta(\mu))] \end{aligned} \quad (3-5)$$

where q_{S/G_0} is the initial state of the observer $G_{\text{obs}}^{S/G}$ of S/G and q'_0 is the initial state of the observer G_{obs}^A of $S \circ A/G$. $\square$

If there exists a sequence μ generated by $S \circ A/G$ such that $\theta(\mu) \in P_o(L(S/G))$, then there is no dump state generated by μ in $S \circ A/G$. Condition (3-5) then requires that the set of all possible states after observing $\theta(\mu)$ should always contain a nonsingleton state or a singleton with a wrong state estimate. Alternately speaking, if $S \circ A/G$ is deadlock-free, then the attackable strong detectability describes that one cannot determine correctly, after a finite number of observations, the current state and subsequent states of $S \circ A/G$ for certain (not

Fig. 7. Plant G and a supervisor S in Example 2. (a) Plant G . (b) Supervisor S .Fig. 8. Attacker A and the observer of S/G in Example 2. (a) Attacker A . (b) Observer of S/G .Fig. 9. Closed-loop system under attack $S \circ A/G$ and the observer of $S \circ A/G$ in Example 2. (a) $S \circ A/G$. (b) Observer of $S \circ A/G$.

all) arbitrarily long trajectories of the system. Different from the aforementioned definition of strong detectability, due to the presence of attack, the violation of strong detectability involves not only the case of a nonsingleton state estimate but also the case of a singleton with an incorrect state estimate. The latter will never happen in a closed-loop system without a sensor attack.

Example 2: A plant G and its supervisor S are depicted in Fig. 7, where $\Sigma_o \setminus \Sigma_{o,p} = \Sigma_o = \Sigma_c = \Sigma = \{a, b, c, d, e\}$. The attacker A and the observer of S/G are shown in Fig. 8. The closed-loop system under attack $S \circ A/G$ and the observer of $S \circ A/G$ are shown in Fig. 9.

In this example, the observer of $S \circ A/G$ is constructed to make state estimations under attack. For the closed-loop system under attack $S \circ A/G$, the set of all possible states after observing bde^* or ade^* contains a singleton with a wrong state estimate. We cannot determine correctly the current and subsequent states after bde^* and ade^* are observed. Therefore, $S \circ A/G$ is attackable with respect to strong detectability. $\square$

Definition 4: Given a plant G , a supervisor S , and a deterministic attacker A , the closed-loop system under attack $S \circ A/G = ((Z \times Y \cup \{d\}) \times X, \Sigma \times \Delta_n, \kappa, (z_0, y_0, x_0), Z \times Y \times X_m)$ is said to be attackable with respect to weak detectability and Σ_o , if S/G is weakly detectable and the following predicate is true:

$$(\forall k \in \mathbb{N}) (\forall \mu \in L(S \circ A/G)) |\theta(\mu)| \geq k \wedge (\forall \mu' \in (\Sigma \times$$

$$\begin{aligned}
\Delta_n^*)[|\theta(\mu')| \geq |X| \wedge \mu\mu' \in L(S \circ A/G) \wedge \theta(\mu\mu') \in \\
P_o(L(S/G)) \Rightarrow (\exists t \leq \mu') [R_{S \circ A/G}(q'_0, \theta(\mu t)) | \\
\{\xi(x_0, \psi(\mu t))\} \neq R_{S/G}(q_{S/G_0}, \theta(\mu t))]]
\end{aligned} \quad (3-6)$$

where q_{S/G_0} is the initial state of the observer $G_{\text{obs}}^{S/G}$ of S/G and q'_0 is the initial state of the observer G_{obs}^A of $S \circ A/G$. $\square$

Given a sequence μ generated by $S \circ A/G$, let μ' be a sufficiently long continuation of μ with $|\theta(\mu')| \geq |X|$ and $\mu\mu' \in L(S \circ A/G)$. If $\theta(\mu\mu') \in P_o(L(S/G))$, then there is no dump state in $S \circ A/G$. Condition (3-6) then requires that there should exist a prefix substring t of μ' , such that the set of all possible states after observing $\theta(\mu t)$ contains a nonsingleton state or a singleton with a wrong state estimate. In other words, the attackable weak detectability describes that we cannot determine correctly, after a finite number of observations, the current state and subsequent states of $S \circ A/G$ for all arbitrarily long trajectories of the system.

So far we have introduced a simple sensor attack model and explained how this attack affects the closed-loop system. However, we have not described what types of sensor attacks can be considered to be *smart*. Next, we will first touch upon the basic properties that a smart sensor attack needs to satisfy, followed by the concepts of AASD and AAWD.

Definition 5 (Attack covertness): Given a deterministic attacker A and a closed-loop system S/G , A is said to be covert if the event insertions launched by A are covert to S , i.e.,

$$\theta(L(A)) \subseteq L(S). \quad (3-7)$$

$\square$

By definition, if the attacker A is covert, then the supervisor will not see any unexpected observation sequences due to the presence of A ; otherwise, the supervisor will see an unexpected observation sequence and interpret this anomaly as the consequence of an attacker. A subsequent remedy action may be accordingly taken by the supervisor, e.g., synchronizing with the plant and the state estimator again, such that the system is detectable again.

Definition 6 (Attack control feasibility): Given a deterministic attacker A and a closed-loop system S/G , A is said to be control-feasible if the language of the closed-loop system under attack is normal [2], [6] with respect to (G, P_o) , i.e.,

$$P_o^{-1}(P_o(\psi(L(S \circ A/G)))) \cap L(G) = \psi(L(S \circ A/G)). \quad (3-8)$$

$\square$

The property of control feasibility of an attacker A is necessary for the observability of the closed-loop behavior of $S \circ A/G$. By imposing normality to achieve observability, one guarantees the existence of the maximal AASD and AAWD models, which will be introduced shortly. In addition, imposing normality can additionally bring certain computational advantages.

Definition 7 (Attack liveness): Given a deterministic attacker A and a closed-loop system S/G , A is said to be attack-live if it does not corrupt the liveness property of the closed-loop system,

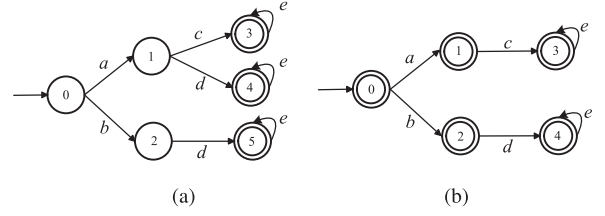


Fig. 10. Plant G and a supervisor S in Example 3. (a) Plant G . (b) Supervisor S .

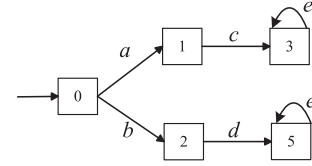


Fig. 11. Observer of S/G in Example 3.

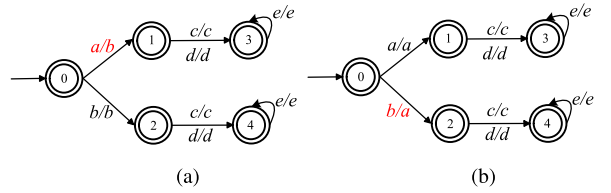


Fig. 12. Attackers A_1 and A_2 in Example 3. (a) Attacker A_1 . (b) Attacker A_2 .

i.e.,

$$\begin{aligned}
(\forall \mu \in L(S \circ A/G)) (\exists (\sigma, v) \in \Sigma \times \Delta_n) \\
\mu(\sigma, v) \in L(S \circ A/G). \quad (3-9)
\end{aligned}$$

$\square$

If an attacker demolishes the attack liveness property, then a deadlock may be detected at the supervisor end.

B. Attack Against Strong Detectability (AASD) Model

Definition 8 (Smart attackability with respect to strong detectability): A closed-loop system S/G is said to be smartly attackable with respect to strong detectability if there exists a nonempty attack model A such that A satisfies the following:

- 1) attack covertness;
- 2) strong detectability damage, i.e., $S \circ A/G$ is attackable with respect to strong detectability under Σ_o ;
- 3) attack control feasibility; and
- 4) attack liveness.

Such a model A is called an AASD of S/G . In addition, A is said to be nonredundant w.r.t. S/G if $L(S \circ A/G) = L(A)$. $\square$

Example 3: A plant G and its supervisor S are shown in Fig. 10, where $\Sigma_o \setminus \Sigma_{o,p} = \Sigma_o = \Sigma_c = \Sigma = \{a, b, c, d, e\}$.

The observer of S/G is shown in Fig. 11. We conclude that S/G is strongly and weakly detectable thanks to Definitions 1 and 2, respectively. In this example, two different attackers A_1 and A_2 , as shown in Fig. 12, are considered.

The composition of $S \circ A_1$ and the closed-loop system under attack $S \circ A_1/G$ are shown in Fig. 13(a) and (b), respectively. At the beginning, both the supervisor S and the attacker A_1 are at

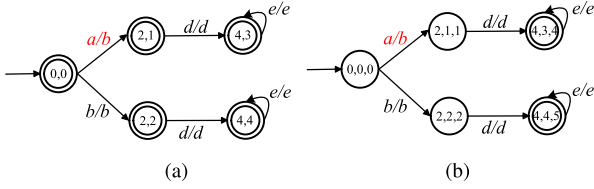


Fig. 13. Attacked supervisor $S \circ A_1$ and the closed-loop system under attack $S \circ A_1/G$ in Example 3. (a) $S \circ A_1$. (b) $S \circ A_1/G$.

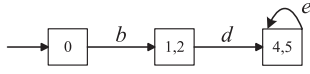


Fig. 14. Observer of $S \circ A_1/G$ in Example 3.

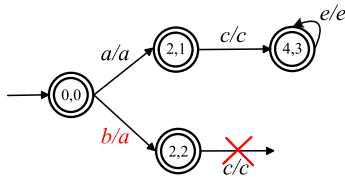


Fig. 15. Attacked supervisor $S \circ A_2$ in Example 3.

their initial states $(0,0)$. Hypothetically, the event a is generated by the plant, which is intercepted by the attacker A_1 . Then, A_1 replaces event a with event b , and sends b to the supervisor S . After that, A_1 updates its state from 0 to 1. After receiving the faked event b from A_1 , the supervisor S moves its state from 0 to 2, and generates a new control command $\gamma = \{d\}$, which is sent to the plant G . At the same time, the state estimator will receive the observation of event b rather than event a . In this case, the resulting state tuple in the composition $S \circ A_1$, due to the reception of event a from the plant G , is $(2,1)$. Other transitions and states can be generated in the same way, and the resulting composition $S \circ A_1$ is shown in Fig. 13(a). Under the supervision of the new attacked supervisor $S \circ A_1$, the closed-loop system under attack $S \circ A_1/G$ is shown in Fig. 13(b).

The observer of the closed-loop system under attack $S \circ A_1/G$ is constructed, as shown in Fig. 14. Clearly, the attacker A_1 is smart, i.e., A_1 is an AASD since it satisfies all the conditions presented in Definition 8.

Then, we consider the attacker A_2 . The attacked supervisor $S \circ A_2$ is shown in Fig. 15. A_2 is not an AASD as it violates the attack liveness property requirement of Definition 8. $\square$

C. Attack Against Weak Detectability (AAWD) Model

Definition 9 (Smart attackability with respect to weak detectability): A closed-loop system S/G is said to be smartly attackable with respect to weak detectability if there exists a nonempty attack model A such that A satisfies the following:

- 1) attack covertness;
- 2) weak detectability damage, i.e., $S \circ A/G$ is attackable with respect to weak detectability under Σ_o ;
- 3) attack control feasibility; and
- 4) attack liveness.

Such a model A is called an AAWD of S/G . In addition, A is said to be nonredundant w.r.t. S/G if $L(S \circ A/G) = L(A)$. $\square$

Theorem 1: Given a plant G and a supervisor S , let $\{A_i | i \in I\}$ be a (possibly infinite) collection of AASD models with respect to (G, S) , where I is an index set possibly infinite. Then, $\bigcup_{i \in I} A_i$ satisfies properties (1)–(4) in Definition 8.

Proof: Properties (1)–(4) in Definition 8 are verified separately.

1) For the covertness, since for any $i \in I$, A_i is an AASD model, implying $\theta(L(A_i)) \subseteq L(S)$. It gives

$$\theta(L(\bigcup_{i \in I} A_i)) = \theta(\bigcup_{i \in I} L(A_i)) = \bigcup_{i \in I} \theta(L(A_i)) \subseteq L(S).$$

In words, the supervisor will not see any unexpected observation sequences from $\bigcup_{i \in I} A_i$.

2) For the strong detectability damage, there exist $s \in L(S \circ (\bigcup_{i \in I} A_i)/G) = L(\bigcup_{i \in I} (S \circ A_i)/G)$ and $h \in I$ such that $s \in L(S \circ A_h/G)$. Since A_h is an AASD model, one necessarily finds $t \in L(S \circ A_h/G) \subseteq L(S \circ (\bigcup_{i \in I} A_i)/G)$ such that $s \leq t$ and the following condition is satisfied:

$$|R_{S \circ A_h/G}(q'_0, \theta(t))| > 1 \vee \{\xi(x_0, \psi(t))\} \neq R_{S/G}(q_{S/G_0}, \theta(t)).$$

3) For the control feasibility, it can be shown straightforwardly, thanks to $L(S \circ (\bigcup_{i \in I} A_i)/G) = \bigcup_{i \in I} L(S \circ A_i)/G$.

4) For the attack liveness, it follows from Definition 7 and the sequential composition $S \circ A$. This completes the proof of the theorem. $\square$

Theorem 2: Given a plant G and a supervisor S , let $\{A_i | i \in I\}$ be a (possibly infinite) collection of AAWD models with respect to (G, S) , where I is an index set possibly infinite. Then, $\bigcup_{i \in I} A_i$ satisfies properties (1)–(4) in Definition 9.

Proof: It follows from Definition 9 and Theorem 1. $\square$

Before closing this section, we state the main problems to be addressed in this article. In the next section, we will first show how to verify the attackable strong detectability and the attackable weak detectability of a closed-loop system under attack.

Problem 1: Given a strongly detectable closed-loop system S/G , compute the supremal nonredundant AASD language of S/G with respect to $\Sigma_{o,p}$. $\square$

Problem 2: Given a weakly detectable closed-loop system S/G , compute the supremal nonredundant AAWD language of S/G with respect to $\Sigma_{o,p}$. $\square$

IV. VERIFICATION OF ATTACKABLE STRONG DETECTABILITY AND ATTACKABLE WEAK DETECTABILITY

Before verifying the attackable strong detectability and attackable weak detectability, we denote the observers of S/G and $S \circ A/G$ by $G_{\text{obs}}^{S/G}$ and G_{obs}^A , respectively, and also denote the sets of all loops in $G_{\text{obs}}^{S/G}$ and G_{obs}^A by Loop and Loop^A , respectively. The set of states in G_{obs}^A that are of singleton is denoted by Q'_s .

Theorem 3: Given a plant G , a supervisor S , and a deterministic attacker A , assume that S/G is strongly detectable and $S \circ A/G$ is deadlock-free. Then, $S \circ A/G$ is attackable with respect to strong detectability under Σ_o if and only if, in $G_{\text{obs}}^{S/G}$

and G_{obs}^A

$$(\exists(q', u) \in \text{Loop}^A)(\exists w \in \Sigma_o^*)[\delta'(q', w) \notin Q'_s] \vee \mathcal{D}_1$$

where $G_{\text{obs}}^{S/G} = (Q_{S/G}, \Sigma_o, \delta_{S/G}, q_{S/G_0})$, $G_{\text{obs}}^A = (Q', \Delta_n, \delta', q'_0)$, and $\mathcal{D}_1$ is a predicate

$$(\exists h \in \{s \in \Sigma_o^* | \delta'(q'_0, s) = q'\})[\delta_{S/G}(q_{S/G_0}, hw)!] \wedge [\delta'(q', w) \neq \delta_{S/G}(q_{S/G_0}, hw)].$$

Proof: (Only if) Suppose that the closed-loop system under attack $S \circ A/G$ is attackable with respect to strong detectability under Σ_o , that is, the current state and subsequent states cannot be determined for certain (not all) trajectories of the system correctly. By Definition 3, let us now suppose *ad absurdum* that the following predicate holds true:

$$(\forall(q', u) \in \text{Loop}^A)(\forall w \in \Sigma_o^*)[\delta'(q', w) \in Q'_s] \wedge \neg \mathcal{D}_1$$

where $\neg \mathcal{D}_1$ is

$$(\forall h \in \{s \in \Sigma_o^* | \delta'(q'_0, s) = q'\})[\delta_{S/G}(q_{S/G_0}, hw) \text{ undefined}] \vee [\delta'(q', w) = \delta_{S/G}(q_{S/G_0}, hw)].$$

Then, the following statements are satisfied.

Since for all $(q', u) \in \text{Loop}^A$ and $w \in \Sigma_o^*$, $[\delta'(q', w) \in Q'_s]$ is true, all the states reachable from q' have the singleton state estimation $\delta'(q', w)$. For all $h \in \{s \in \Sigma_o^* | \delta'(q'_0, s) = q'\}$ and $w \in \Sigma_o^*$, $[\delta_{S/G}(q_{S/G_0}, hw) \text{ undefined}]$ is false. Therefore, the truth of predicate $\neg \mathcal{D}_1$ means that, for all $h \in \{s \in \Sigma_o^* | \delta'(q'_0, s) = q'\}$ and $w \in \Sigma_o^*$, $[\delta'(q', w) = \delta_{S/G}(q_{S/G_0}, hw)]$ is true. Given a loop (q', u) in Loop^A , there exists a consistent loop $(\delta_{S/G}(q_{S/G_0}, h), u)$ in Loop such that $R_{S \circ A/G}(q'_0, hu^*w) = R_{S/G}(q_{S/G_0}, hu^*w)$, signifying that the current state and subsequent states can be determined correctly. Therefore, $S \circ A/G$ is not attackable with respect to strong detectability, which contradicts that $S \circ A/G$ is attackable with respect to strong detectability under Σ_o .

(If) Suppose that the following condition holds:

$$(\exists(q', u) \in \text{Loop}^A)(\exists w \in \Sigma_o^*)[\delta'(q', w) \notin Q'_s] \vee \mathcal{D}_1.$$

There exists a state reachable from a loop of G_{obs}^A such that the cardinality of this state is not equal to 1, or if the cardinality of this state is equal to 1, then the state is not consistent with $G_{\text{obs}}^{S/G}$. Therefore, there exists an arbitrarily long sequence $\mu \in L(S \circ A/G)$ with $\theta(\mu) = hu^*w$, such that $R_{S \circ A/G}(q'_0, hu^*w)$ contains a nonsingleton state or a singleton with a wrong state estimate. By Definition 3, the closed-loop system $S \circ A/G$ is attackable with respect to strong detectability under Σ_o . $\square$

Theorem 3 tells us that the closed-loop system $S \circ A/G$ is attackable w.r.t. strong detectability under Σ_o iff there exists a state reachable from a loop in G_{obs}^A and this state is not in Q'_s or is not consistent with $G_{\text{obs}}^{S/G}$. To verify the attackable strong detectability of a deadlock-free closed-loop system $S \circ A/G$, the following procedure is presented.

Note that, in Procedure 1, Step 3) constructs the observer of $S \circ A/G$, using the method similar to the observer construction of G_{obs} in [9]. Step 4.a) finds a state reachable from a loop in G_{obs}^A and the cardinality of this state is not equal to 1. Based on Theorem 3, if the above state exists, then $S \circ A/G$ is

Procedure 1: Verification of Attackable Strong Detectability.

Input: a plant $G = (X, \Sigma, \xi, x_0, X_m)$, a supervisor $S = (Z, \Sigma, \tau, z_0, Z)$, and an attacker $A = (Y, \Sigma \times \Delta_n, \eta, y_0, Y)$.

- 1) Construct the observer $G_{\text{obs}}^{S/G}$ of S/G .
 - 2) Find $S \circ A/G = (Z \times Y \times X, \Sigma \times \Delta_n, \kappa, (z_0, y_0, x_0), Z \times Y \times X_m)$.
 - 3) Construct the observer G_{obs}^A of $S \circ A/G$.
 - a) Define $q' = UR_{S/G}(\{x_0\})$ and set $Q' = q'$.
 - b) For each $D \in Q'$ and $(\sigma, v) \in \Sigma \times \Delta_n$, define $\delta'(D, v) = R_{S \circ A/G}(D, v)$.
 - c) Repeat Step 3.b) until the entire accessible part of G_{obs}^A has been constructed.
 - 4) For each $(q', u) \in \text{Loop}^A$,
 - a) if $(\exists w \in \Sigma_o^*)[\delta'(q', w)] > 1$, go to Step 6); otherwise, continue.
 - b) if $(\exists w \in \Sigma_o^*)(\exists h \in \Sigma_o^*)[\delta'(q'_0, h) = q'] \wedge [\delta_{S/G}(q_{S/G_0}, hw)!] \wedge [\delta'(q', w) \neq \delta_{S/G}(q_{S/G_0}, hw)]$, go to Step 6); otherwise, continue.
 - 5) Output: $S \circ A/G$ is not attackable with respect to strong detectability; go to Step 7).
 - 6) Output: $S \circ A/G$ is attackable with respect to strong detectability.
 - 7) Exit. $\square$
-

attackable with respect to strong detectability. Step 4.b) finds a state reachable from a loop in G_{obs}^A that, although the cardinality of this state is 1, is not consistent with $G_{\text{obs}}^{S/G}$. If the above state exists, then $S \circ A/G$ is also attackable with respect to strong detectability.

Theorem 4: Given a plant G , a supervisor S , and a deterministic attacker A , assume that S/G is weakly detectable and $S \circ A/G$ is deadlock-free. Then, $S \circ A/G$ is attackable with respect to weak detectability under Σ_o if and only if, in $G_{\text{obs}}^{S/G}$ and G_{obs}^A

$$(\forall(q', u) \in \text{Loop}^A)(\exists w \in \bar{u})[\delta'(q', w) \notin Q'_s] \vee \mathcal{D}_2$$

where $G_{\text{obs}}^{S/G} = (Q_{S/G}, \Sigma_o, \delta_{S/G}, q_{S/G_0})$, $G_{\text{obs}}^A = (Q', \Delta_n, \delta', q'_0)$, and $\mathcal{D}_2$ is a predicate

$$(\exists h \in \{s \in \Sigma_o^* | \delta'(q'_0, s) = q'\})[\delta_{S/G}(q_{S/G_0}, hw)!] \wedge [\delta'(q', w) \neq \delta_{S/G}(q_{S/G_0}, hw)].$$

Proof: (Only if) Suppose that the closed-loop system under attack $S \circ A/G$ is attackable with respect to weak detectability under Σ_o , that is, the current state and subsequent states cannot be determined for all arbitrarily long trajectories of the system correctly. By Definition 4, let us now suppose *ad absurdum* that the following predicate holds true:

$$(\exists(q', u) \in \text{Loop}^A)(\forall w \in \bar{u})[\delta'(q', w) \in Q'_s] \wedge \neg \mathcal{D}_2$$

where $\neg \mathcal{D}_2$ is

$$(\forall h \in \{s \in \Sigma_o^* | \delta'(q'_0, s) = q'\})[\delta_{S/G}(q_{S/G_0}, hw) \text{ undefined}] \vee [\delta'(q', w) = \delta_{S/G}(q_{S/G_0}, hw)].$$

Then, the following conditions are satisfied.

There exists a loop $(q', u) \in \text{Loop}^A$, for all $w \in \bar{u}$, such that $[\delta'(q', w) \in Q'_s]$ is true, implying that there exists a loop (q', u) in G_{obs}^A such that the cardinality of each state $(\delta'(q', w))$ is equal to 1. For all $w \in \bar{u}$ and $h \in \{s \in \Sigma_o^* | \delta'(q'_0, s) = q'\}$, $[\delta_{S/G}(q_{S/G_0}, hw)$ undefined] is false. Therefore, the truth of predicate $\neg \mathcal{D}_2$ means that, for all $w \in \bar{u}$ and $h \in \{s \in \Sigma_o^* | \delta'(q'_0, s) = q'\}$, $[\delta'(q', w) = \delta_{S/G}(q_{S/G_0}, hw)]$ is true. For the loop (q', u) , there exists a consistent loop $(\delta_{S/G}(q_{S/G_0}, h), u)$ in Loop such that $R_{S \circ A/G}(q'_0, hu^*w) = R_{S/G}(q_{S/G_0}, hu^*w)$, indicating that the current state and subsequent states can be determined correctly. The facts indicate that the closed-loop system $S \circ A/G$ is not attackable with respect to weak detectability, which contradicts that the closed-loop system $S \circ A/G$ is attackable with respect to weak detectability under Σ_o .

(If) Suppose that the following condition holds:

$$(\forall (q', u) \in \text{Loop}^A)(\exists w \in \bar{u})[\delta'(q', w) \notin Q'_s] \vee \mathcal{D}_2.$$

There exist some states in each loop of G_{obs}^A such that the cardinality of these states is not equal to 1, or if the cardinality of these states is equal to 1, they are not consistent with $G_{\text{obs}}^{S/G}$. Therefore, for any arbitrarily long sequence $\mu \in L(S \circ A/G)$ with $\theta(\mu) = hu^*$, there always exists a prefix substring w of u such that $R_{S \circ A/G}(q'_0, hu^*w)$ contains a nonsingleton state or a singleton with a wrong state estimate. Thus, $S \circ A/G$ is attackable with respect to weak detectability under Σ_o due to Definition 4. $\square$

Theorem 4 indicates that the closed-loop system $S \circ A/G$ is attackable with respect to weak detectability under Σ_o iff there does not exist a loop in G_{obs}^A that is entirely within Q'_s and is consistent with $G_{\text{obs}}^{S/G}$. Note that $\mathcal{D}_1$ in Theorem 3 is the same expression as $\mathcal{D}_2$ in Theorem 4. To verify the attackable weak detectability of a deadlock-free closed-loop system $S \circ A/G$, the following procedure is proposed.

In Procedure 2, Step 5) checks whether there is a state with a cardinality not equal to 1 in each loop of G_{obs}^A . Based on Theorem 4, if the above states exist, then $S \circ A/G$ is attackable with respect to weak detectability. Step 7) checks whether each loop of G_{obs}^A is not consistent with $G_{\text{obs}}^{S/G}$. If each loop is not consistent with $G_{\text{obs}}^{S/G}$, then $S \circ A/G$ is attackable with respect to weak detectability.

Complexity analysis: Given a plant G , a supervisor S , and a sensor attacker A , the number of states in $S \circ A/G$ is upper bounded by $|X||S||A|$. The complexity of constructing the observer of $S \circ A/G$ is $\mathcal{O}(2^{|X||S||A|})$, and the complexity of constructing the observer of S/G is $\mathcal{O}(2^{|X||S|})$. In Procedures 1 and 2, the computational complexity is $\mathcal{O}(2^{2|X||S||A|+|X||S||\Sigma||\Delta_n|})$, which is exponential with respect to $|X|$, $|S|$, and $|A|$ on the premise that Δ_n is given.

Example 4: A plant G and its supervisor S are depicted in Fig. 16, where $\Sigma_o \setminus \Sigma_{o,p} = \Sigma_o = \Sigma_c = \Sigma = \{a, b, c, d, e\}$. The observer $G_{\text{obs}}^{S/G}$ of S/G is visualized in Fig. 17. S/G is strongly detectable and weakly detectable. The attacker A and the closed-loop system under attack $S \circ A/G$ are shown in Fig. 18. The observer G_{obs}^A of $S \circ A/G$ is portrayed in Fig. 19.

Procedure 2: Verification of Attackable Weak Detectability.

Input: a plant $G = (X, \Sigma, \xi, x_0, X_m)$, a supervisor $S = (Z, \Sigma, \tau, z_0, Z)$, and an attacker $A = (Y, \Sigma \times \Delta_n, \eta, y_0, Y)$.

- 1) Construct the observer $G_{\text{obs}}^{S/G}$ of S/G .
- 2) Compute $S \circ A/G = (Z \times Y \times X, \Sigma \times \Delta_n, \kappa, (z_0, y_0, x_0), Z \times Y \times X_m)$.
- 3) Construct the observer G_{obs}^A of $S \circ A/G$.
 - a) Define $q' = UR_{S/G}(\{x_0\})$; set $Q' = q'$.
 - b) For each $D \in Q'$ and $(\sigma, v) \in \Sigma \times \Delta_n$, define $\delta'(D, v) = R_{S \circ A/G}(D, v)$.
 - c) Repeat Step 3.b) until the entire accessible part of G_{obs}^A has been constructed.
- 4) Let $i, j = 1$.
- 5) For each $(q', u) \in \text{Loop}^A$, if $((\exists w \in \bar{u})|\delta'(q', w)| > 1)$, set $i := i \times 1$; otherwise, set $i := 0$.
- 6) If $(i = 1)$, go to Step 10); otherwise, continue.
- 7) For each $(q', u) \in \text{Loop}^A$,

if $(\exists w \in \bar{u})(\exists h \in \Sigma_o^*)[\delta'(q'_0, h) = q'] \wedge [\delta_{S/G}(q_{S/G_0}, hw)!] \wedge [\delta'(q', w) \neq \delta_{S/G}(q_{S/G_0}, hw)]$, set $j := j \times 1$; otherwise, set $j := 0$.
- 8) If $(j = 1)$, go to Step 10); otherwise, continue.
- 9) Output: $S \circ A/G$ is not attackable with respect to weak detectability; go to Step 11).
- 10) Output: $S \circ A/G$ is attackable with respect to weak detectability.
- 11) Exit. $\square$

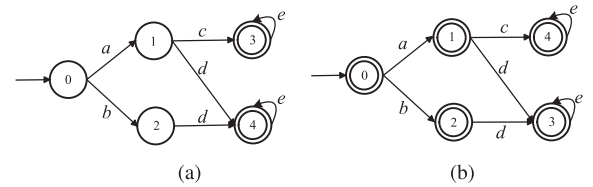


Fig. 16. Plant G and a supervisor S in Example 4. (a) Plant G . (b) Supervisor S .

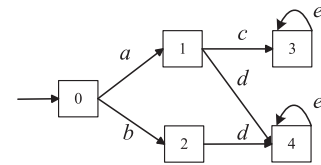


Fig. 17. Observer $G_{\text{obs}}^{S/G}$ of S/G in Example 4.

In this example, any state reachable from any loop in G_{obs}^A is in Q'_s and is consistent with $G_{\text{obs}}^{S/G}$, i.e., for the closed-loop system under attack $S \circ A/G$, we can always make a correct state estimation (x_4) after observing bde^* . Therefore, $S \circ A/G$ is neither attackable with respect to strong detectability nor attackable with respect to weak detectability. $\square$

Example 5: Consider a plant G and its supervisor S in Fig. 20 with $\Sigma_o \setminus \Sigma_{o,p} = \Sigma_o = \Sigma_c = \{a, b, d, e\}$. The observer $G_{\text{obs}}^{S/G}$ of S/G is visualized in Fig. 21. The attacker A and the

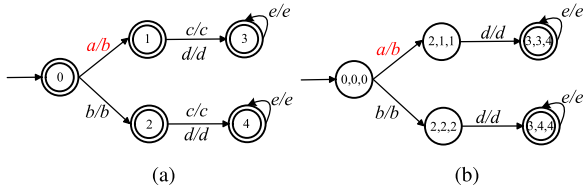


Fig. 18. Attacker A and the closed-loop system under attack $S \circ A/G$ in Example 4. (a) Attacker A . (b) $S \circ A/G$.

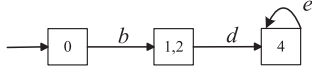


Fig. 19. Observer G_{obs}^A of $S \circ A/G$ in Example 4.

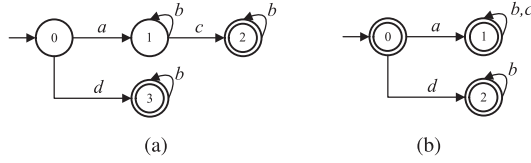


Fig. 20. Plant G and a supervisor S in Example 5. (a) Plant G . (b) Supervisor S .

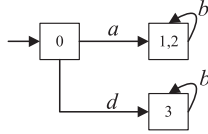


Fig. 21. Observer $G_{\text{obs}}^{S/G}$ of S/G in Example 5.

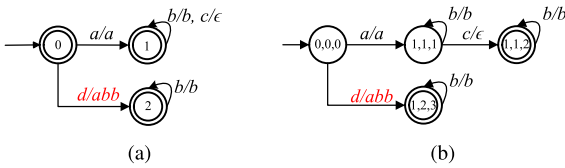


Fig. 22. Attacker A and the closed-loop system under attack $S \circ A/G$ in Example 5. (a) Attacker A . (b) $S \circ A/G$.

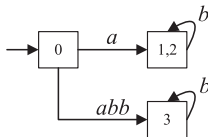


Fig. 23. Observer G_{obs}^A of $S \circ A/G$ in Example 5.

closed-loop system under attack $S \circ A/G$ are shown in Fig. 22. The observer G_{obs}^A of $S \circ A/G$ is portrayed in Fig. 23.

For the closed-loop system S/G , after db^* is observed, the state of the plant is estimated to be x_3 , and the current state and subsequent states can be determined, implying that S/G is weakly detectable. For the closed-loop system under attack $S \circ A/G$, the set of all possible states after observing ab^* is a nonsingleton state or a singleton with a wrong state estimate. In this example, there does not exist a loop in G_{obs}^A that is entirely within Q'_s and is consistent with $G_{\text{obs}}^{S/G}$. Therefore, $S \circ A/G$ is attackable with respect to weak detectability. $\square$

Procedure 3: Supremal AAWD Synthesis.

Input: a plant $G = (X, \Sigma, \xi, x_0, X_m)$, a supervisor $S = (Z, \Sigma, \tau, z_0, Z)$ (S/G is weakly detectable), and $\Sigma_{o,p}$.

- 1) Construct a single-state automaton $A_0 = (Y, \Sigma \times \Delta_n, \eta, y_0, Y)$, where $Y = \{y_0\}$ and the transition map η encodes the transitions labeled by the events in $((\Sigma_o \setminus \Sigma_{o,p}) \times \Delta_n) \cup (\Sigma_{o,p} \times \Sigma_{o,p}) \cup (\Sigma_{uo} \times \{\epsilon\})$, denoting all observable event alterations that the attacker can make.
- 2) Compute the sequential composition $S \circ A_0 = (Z \times Y \cup \{d\}, \Sigma \times \Delta_n, \zeta, (z_0, y_0), Z \times Y \cup \{d\})$.
- 3) Find an automaton $B = (S \circ A_0)/G = ((Z \times Y \cup \{d\}) \times X, \Sigma \times \Delta_n, \kappa, (z_0, y_0, x_0), Z \times Y \times X_m)$.
- 4) Compute B_1 via removing the dump states and dead states from B according to controllability and normality [1].
- 5) Construct the observer $G_{\text{obs}}^{S/G}$ of S/G .
- 6) Construct the observer G_{obs}^A via the subset construction on B_1 (according to Procedure 1).
- 7) Find $Loop$, i.e., the set of loops in $G_{\text{obs}}^{S/G}$.
- 8) Find $Loop^A$, i.e., the set of loops in G_{obs}^A .
- 9) For each $(q', u) \in Loop^A$, if $([(\forall w \in \bar{u})|\delta'(q', w)| = 1] \wedge [(\forall h \in \Sigma_o^*)[\delta'(q'_0, h) = q'] \wedge [\delta_{S/G}(q_{S/G_0}, hw)!] \wedge [\delta'(q', w) = \delta_{S/G}(q_{S/G_0}, hw)]])$, remove events starting from $\mu^\dagger$ in B_1 such that $L(B_1) = L(B_1) - \mu$, where $\{\mu \in L(B_1) | (\exists (z, y, x) \in (Z \times Y \times q'))[\kappa((z_0, y_0, x_0), \mu) = (z, y, x)] \wedge [\theta(\mu) = hu^*] \wedge [\delta'(q'_0, \theta(\mu)) = q']\}$ according to liveness, controllability, and normality [1], and go to Step 6); otherwise, continue.
- 10) Let $A_w = B_1$.
- 11) Output: the supremal AAWD model is the prefix closure of A_w .
- 12) Exit. $\square$

V. SYNTHESIS OF THE SMART SENSOR ATTACK FOR TAMPERING STRONG DETECTABILITY AND WEAK DETECTABILITY

This section shows the conditions under which the supremal AAWD and supremal AASD exist and are computable. To solve Problems 1 and 2, the following synthesis procedures are presented.

In Procedure 3, Step 2) is to determine all possible observation alterations on supervisor S . Step 3) describes the closed-loop language under the influence of A_0 . The computation carried out in Step 4) ensures that the closed-loop language is deadlock-free, normal, and controllable. Since (G, S) is deadlock-free, if the system is in a deadlock state after launching attacks, even though the strong or weak detectability is tempered, then we will be aware that the attack has occurred at the same time. Steps 5) and 6) construct the observer of S/G and the observer of B_1 , respectively. Steps 7)–9) search for the loops in G_{obs}^A that are

completely contained in Q'_s and are consistent with $G_{\text{obs}}^{S/G}$. If existing, we then remove events in B_1 to prevent G_{obs}^A from generating these loops. Clearly, each step in Procedure 3 terminates in a finite time, so does Procedure 3. The prefix closure of A_w in Step 11), if not empty, is the supremal nonredundant AAWD model.

Proposition 10: If the output A_w in Procedure 3 is not empty, then its prefix closure is the supremal nonredundant AAWD of (G, S) with respect to $\Sigma_{o,p}$; otherwise, there is no AAWD of (G, S) with respect to $\Sigma_{o,p}$.

Proof: We first show that the prefix closure A_* of A_w is an AAWD, i.e., A_* satisfies properties (1)–(4) in Definition 9. By the definition of $S \circ A_0$, the dump state d is a deadlock state. By the definition of B in Steps 3) and 4) of Procedure 3, no state in B_1 (thus in A_*) contains d as an element, i.e., $\theta(L(A_*)) = \theta(L(B_1)) \subseteq L(S)$, which completes the proof of property (1). Step 4) computes a normal deadlock-free sublanguage [1], which ensures that properties (3) and (4) are satisfied.

Further, for all $\mu \in L(S \circ A_*/G)$, and any its sufficiently long continuation $\mu' \in L(S \circ A_*/G)$, the following condition is satisfied:

$$(\exists t \leq \mu') \theta(\mu t) \in P_o(L(S/G)) \wedge [|R_{B_1}(q_0, \theta(\mu t))| > 1 \vee \{\xi(x_0, \psi(\mu t))\} \neq R_{S/G}(q_0, \theta(\mu t))].$$

Therefore, Steps 7)–9) ensure that property (2) is satisfied by Definition 4 and Theorem 4.

To show that A_* is nonredundant w.r.t. (G, S) , we then only need to verify if $L(S \circ A_*/G) = L(A_*)$ is true. By $L(S \circ A_*/G) \subseteq L(A_*)$, we need to show $L(S \circ A_*/G) \supseteq L(A_*)$. This is in fact true, thanks to $L(A_*) = L(B_1) \subseteq L(S \circ B_1/G)$.

Since A_* is a nonredundant AAWD model, to show that A_* is the supremal nonredundant AAWD model, we only need to show that for all nonredundant AAWD models $\hat{A}$, we have $L(\hat{A}) = L(S \circ \hat{A}/G) \subseteq L(S \circ A_*/G) = L(A_*)$. Clearly, $L(S \circ \hat{A}) \subseteq L(S \circ A_0)$. Thus, it gives $L(S \circ \hat{A}/G) \subseteq L(B)$. Owing to properties (2)–(4) in Definition 9 and the construction of A_* , we have $L(S \circ \hat{A}/G) \subseteq L(B_1) = L(A_*) = L(S \circ A_*/G)$. Thus, if the output A_w in Procedure 3 is not empty, then its prefix closure A_* is the supremal nonredundant AAWD of (G, S) with respect to $\Sigma_{o,p}$. $\square$

In Procedure 4, Steps 7) and 8) find effective attack paths such that there exist some states reachable from any loop in G_{obs}^A that are not in Q'_s or are not consistent with $G_{\text{obs}}^{S/G}$. Clearly, each step in Procedure 4 terminates in a finite time. The prefix closure of A_s in Step 9), if not empty, is the supremal nonredundant AASD model.

Proposition 11: If the output A_s in Procedure 4 is not empty, then its prefix closure is the supremal nonredundant AASD of (G, S) with respect to $\Sigma_{o,p}$; otherwise, there is no AASD of (G, S) with respect to $\Sigma_{o,p}$.

Proof: It follows from Theorem 3 and Proposition 10. $\square$

Complexity analysis: The computational complexity of Procedures 3 or 4 can be roughly estimated as follows. The complexity of Step 4) in Procedures 3 and 4 is $\mathcal{O}(2^{3|X||S|}|\Sigma||\Delta_n|)$ [1]. The number of states in B_1 is upper bounded by $|X|(|S||A_0| +$

Procedure 4: Supremal AASD Synthesis.

Input: a plant $G = (X, \Sigma, \xi, x_0, X_m)$, a supervisor $S = (Z, \Sigma, \tau, z_0, Z)$ (S/G is strongly detectable), and $\Sigma_{o,p}$.

- 1) Construct a single-state automaton $A_0 = (Y, \Sigma \times \Delta_n, \eta, y_0, Y)$, where $Y = \{y_0\}$ and the transition map η encodes the transitions labeled by the events in $((\Sigma_o \setminus \Sigma_{o,p}) \times \Delta_n) \cup (\Sigma_{o,p} \times \Sigma_{o,p}) \cup (\Sigma_{uo} \times \{\epsilon\})$, denoting all observable event alterations that the attacker can make.
 - 2) Compute the sequential composition $S \circ A_0 = (Z \times Y \cup \{d\}, \Sigma \times \Delta_n, \zeta, (z_0, y_0), Z \times Y \cup \{d\})$.
 - 3) Find an automaton $B = (S \circ A_0)/G = ((Z \times Y \cup \{d\}) \times X, \Sigma \times \Delta_n, \kappa, (z_0, y_0, x_0), Z \times Y \times X_m)$.
 - 4) Compute B_1 via removing dump states and dead states from B according to controllability and normality [1].
 - 5) Construct the observer $G_{\text{obs}}^{S/G}$ of S/G .
 - 6) Construct the observer G_{obs}^A via the subset construction on B_1 (according to Procedure 1).
 - 7) Find *Loop*, i.e., the set of loops in $G_{\text{obs}}^{S/G}$, and find *Loop*^A, i.e., the set of loops in G_{obs}^A .
 - 8) For each $(q', u) \in \text{Loop}^A$, if $((\exists w \in \Sigma_o^*)[\delta'(q', w)] \neq 1] \vee [(\exists h \in \Sigma_o^*)[\delta'(q'_0, h) = q'] \wedge [\delta_{S/G}(q_{S/G_0}, hw)] \wedge [\delta'(q', w) \neq \delta_{S/G}(q_{S/G_0}, hw)]]$, let $A_s = B_1$ and go to Step 9); otherwise, go to Step 10).
 - 9) Output: the supremal AASD model is the prefix closure of A_s ; go to Step 11).
 - 10) Output: the supremal AASD model does not exist.
 - 11) Exit. $\square$
-

1) = $|X|(|S| + 1) \approx |X||S|$. The complexity of constructing the observer of B_1 is $\mathcal{O}(2^{|X||S|})$. The overall complexity of constructing A_w is

$$\mathcal{O}(2^{3|X||S|}|\Sigma||\Delta_n| + 2^{6|X||S|}|\Sigma|^2|\Delta_n|^2).$$

By

$$|\Delta_n| = \frac{|\Sigma|^{n+1} - 1}{|\Sigma| - 1}$$

and $|\Sigma| \approx |\Sigma| - 1$, it becomes

$$\begin{aligned} &\mathcal{O}(2^{3|X||S|}|\Sigma||\Delta_n| + 2^{6|X||S|}|\Sigma|^2|\Delta_n|^2) \\ &= \mathcal{O}(2^{3|X||S|}|\Sigma|^{n+1} + 2^{6|X||S|}|\Sigma|^{2n+2}) \end{aligned}$$

which is exponential with respect to $|X|$ and $|S|$ on the premise that n is given. The overall complexity of constructing A_s is

$$\mathcal{O}(2^{3|X||S|+1}|\Sigma||\Delta_n|) = \mathcal{O}\left(2^{3|X||S|+1}|\Sigma|\frac{|\Sigma|^{n+1} - 1}{|\Sigma| - 1}\right).$$

Considering $|\Sigma| \approx |\Sigma| - 1$, it becomes

$$\mathcal{O}(2^{3|X||S|+1}|\Sigma||\Delta_n|) = \mathcal{O}(2^{3|X||S|+1}|\Sigma|^{n+1})$$

which is exponential with respect to $|X|$ and $|S|$ on the premise that n is given. As an illustration, we apply Procedures 3 and 4 to the plant G and the supervisor S in Example 6.

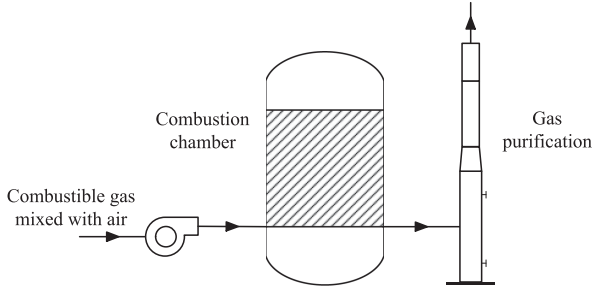
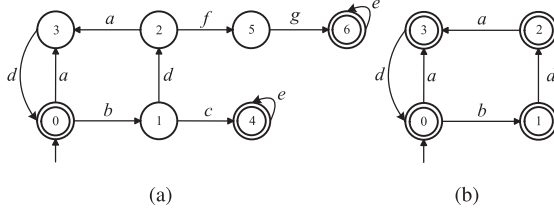


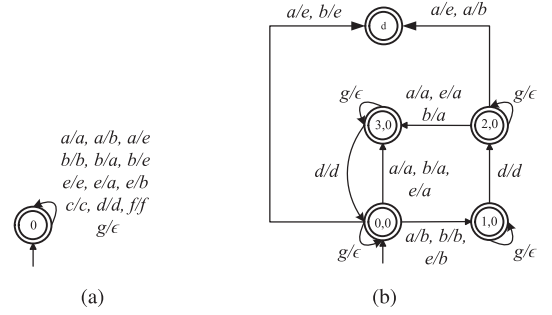
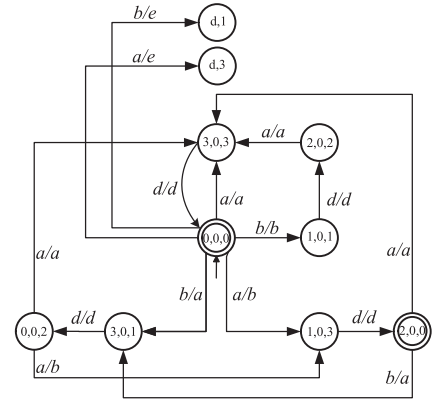
Fig. 24. Waste calciner system.

Fig. 25. Plant G and a supervisor S in Example 6. (a) Plant G . (b) Supervisor S .TABLE I
EVENTS OF PLANT G SHOWN IN FIG. 25(A)

Event	Meaning
a	High
b	Normal
c	Dust removal
d	Heating
e	Standard gas emission
f	Gas denitration
g	Active carbon absorption

Example 6: Let us go through a simple waste calciner example, as depicted in Fig. 24, which consists of a combustible gas mixed with air supply, a combustion chamber, and a gas purification. The combustible gas mixed with air heats the combustion chamber through the blower. The waste is then incinerated in the combustion chamber. Finally, the waste gas generated during the incineration process will be purified (including gas denitration, dust removal, active carbon absorption, and standard gas emission). The temperature in combustion chamber can be measured, whose value can trigger some predefined events, denoting the temperature conditions: high and normal. The administrator will correctly estimate the current load demand of the system by recording the heat output of the combustion chamber (i.e., high temperature and normal temperature) to control the supply of combustible gas mixed with air. If an attacker can intercept and alter the temperature conditions, then the administrator will not be able to accurately estimate the current load requirements of the system, and catastrophic problems may occur.

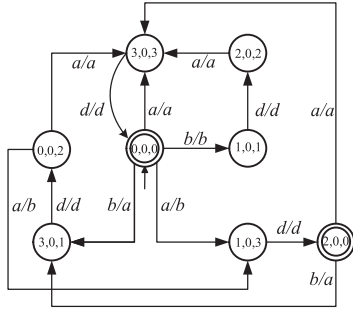
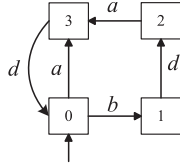
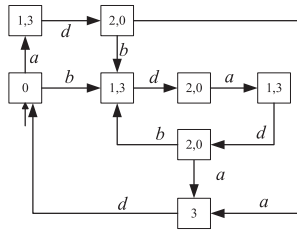
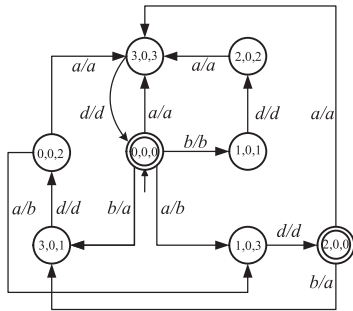
We construct a simple discrete-event model for the system G and a supervisor S , as shown in Fig. 25, where the meanings of the events are shown in Table I. In this example, considering that the combustion chamber temperature will naturally drop at high temperature or normal temperature, when the combustion

Fig. 26. Attacker A_0 and sequential composition $S \circ A_0$ in Example 6. (a) Attacker A_0 . (b) Sequential Composition $S \circ A_0$.Fig. 27. Finite-state automaton B in Example 6.

chamber temperature is high or normal, it will be heated. The combustion chamber temperature may be high after heating or the combustion chamber temperature may naturally drop to normal after heating. However, if the combustion chamber was a normal temperature before, then it will be of high temperature after heating. The supervisor only realizes the heating control of the combustion chamber in this example, and does not perform dust removal and gas denitration. Events c , d , and f are controllable, i.e., $\Sigma_c = \{c, d, f\}$ and the event g is unobservable, i.e., $\Sigma_o = \{a, b, c, d, e, f\}$. Assume that $\Sigma_{o,p} = \{c, d, f\}$ and the length of any observable sequence that the attacker can insert is no more than 1, i.e., $n = 1$.

The finite-state automaton A_0 and the sequential composition $S \circ A_0$ are depicted in Fig. 26. The finite-state automaton $B := (S \circ A_0)/G$ is shown in Fig. 27 and B_1 is visualized in Fig. 28 by stripping away transitions b/e and a/e to eliminate the dump states and dead states in B .

The observer $G_{\text{obs}}^{S/G}$ of S/G and the observer G_{obs}^A of B_1 are shown in Figs. 29 and 30, respectively. S/G is strongly detectable and weakly detectable. We can check that there exists at least one state that the cardinality of this state is not equal to 1 in each loop of G_{obs}^A , i.e., there do not exist loops in G_{obs}^A that are completely contained within Q'_s and are consistent with $G_{\text{obs}}^{S/G}$. Consequently, we have $A_w = B_1$ and the prefix closure of A_w satisfies properties (1)–(4) in Definition 9, with A_w being depicted in Fig. 31. By Proposition 10, the supremal

Fig. 28. Finite-state automaton B_1 in Example 6.Fig. 29. Observer $G_{\text{obs}}^{S/G}$ of S/G in Example 6.Fig. 30. Observer G_{obs}^A of B_1 in Example 6.Fig. 31. Finite-state automaton $A_w(A_s)$ in Example 6.

nonredundant AAWD model is the prefix closure of A_w . We can check that there exist some states [see, e.g., (2,0) and (1,3)] reachable from some loops in G_{obs}^A that are not in Q'_s . As a result, one finds $A_s = B_1$ and the prefix closure of A_w satisfies properties (1)–(4) in Definition 8, with A_s being depicted in Fig. 31. By Proposition 11, the supremal nonredundant AASD model is the prefix closure of A_s . $\square$

VI. CONCLUSION

In this article, we have first introduced the concepts of attackable strong detectability and attackable weak detectability of a closed-loop system under attacks. An attacker is modeled

as a transducer. Then, the notions of AASD and AAWD are presented. Necessary and sufficient conditions for the two types of attackable detectability are derived. Procedure 1 is presented to verify the condition of the attackable strong detectability, and Procedure 2 is to verify the condition of the attackable weak detectability. We have shown that the supremal nonredundant AASD model and supremal nonredundant AAWD model exist and are computable as long as the plant model G and the supervisor S are finitely representable, i.e., their languages are regular. To move forward, we will address the problems of supervisory control against AASD and AAWD.

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